

Embedded systems exp 27

```
#include <systemc.h>
```

```
SC_MODULE(binary_semaphore) {
```

```
    int sem;
```

```
    sc_event ev;
```

```
void task1() {
```

```
    while(true) {
```

```
        wait(1, SC_SEC);
```

```
        if(sem == 1) {
```

```
            sem = 0;
```

```
            cout << "Task1 Entered Critical Section at "
```

```
                << sc_time_stamp() << endl;
```

```
            wait(2, SC_SEC);
```

```
            sem = 1;
```

```
            ev.notify();
```

```
        }
```

```
    else {
```

```
        wait(ev);
```

```
    }
```

```
}
```

```
}
```

```
void task2() {
```

```
    while(true) {
```

```
        wait(2, SC_SEC);
```

```
        if(sem == 1) {
```

```
            sem = 0;
```

```

        cout << "Task2 Entered Critical Section at "
              << sc_time_stamp() << endl;

        wait(1, SC_SEC);

        sem = 1;
        ev.notify();
    }
    else {
        wait(ev);
    }
}

}

SC_CTOR(binary_semaphore) {
    sem = 1;

    SC_THREAD(task1);
    SC_THREAD(task2);
}

};

int sc_main(int argc, char* argv[]) {
    binary_semaphore obj("Binary_Semaphore");

    sc_start(10, SC_SEC);

    return 0;
}

```

Output:

```
[2026-08-29 05:35:26 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim  
Compile done. Starting run...
```

```
SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
```

```
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```

```
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```

```
Task1 Entered Critical Section at 1 s
```

```
Task1 Entered Critical Section at 4 s
```

```
Task1 Entered Critical Section at 7 s
```

```
Done
```
